EFY TESTED
PREETI BHAKTA

VEHICLE TRACKING SYSTEM
Based On GPS And GSM

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Presented here is an Arduino-based system for vehicle tracking using global positioning system (GPS) and global system for mobile communication (GSM) modules. GSM modem with a SIM card used here uses the communication technique of a regular cellphone. The system can be installed or hidden in your vehicle at a suitable location. After installing this circuit, you can easily track your stolen vehicle using a mobile phone. The author's prototype is shown in Fig. 1.

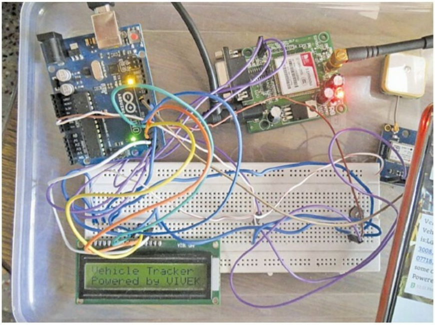

Fig. 1: Author's prototype

Circuit and working

Circuit diagram of the GPS- and GSM-based vehicle tracking system is shown in Fig. 2. It is built around Arduino Uno board (BOARD1), 16x2 LCD (LCD1), GPS module, GSM module and a pot-meter (VR1). Working of the circuit is very simple. To operate the circuit, you need to follow the steps given below:

1. First, install Arduino IDE 1.8.3 in your PC. Add GPRS_Shield_Arduino's header file into the library of Arduino IDE. Download the file from https://github.com/Seeed-Studio/GPRS_SIM900
2. Include or add the user's mobile number (on which you want to receive the GPS location messages) in the source code (vehicle_code.ino). Compile and upload the code onto the Arduino board using Arduino IDE
3. Once the code is burnt, make connections as shown in the circuit diagram.
4. Insert a valid SIM card in the GSM module connected with the Arduino board. Make sure that this SIM has necessary balance to send alert messages to the user's mobile phone.

Construction and testing

An actual-size PCB layout of GPS- and GSM-based vehicle tracking system is shown in Fig. 3 and its components layout in Fig. 4.

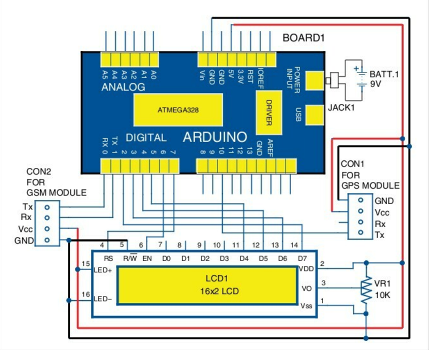

Fig. 2: Circuit diagram of GPS- and GSM-based vehicle tracking system